

ΦΕΒ21
 (4) (7772): 100W22
 (2) ✓

ΚΑΤΟΥΦΗ ΤΩ
 ΚΑΛΩΔΙΟΥ.

$$\rho_1 = \sigma \omega = \sqrt{\kappa - 1} =$$

$$\xi = \xi_r \cdot \xi_\theta$$

αν $\xi_r \in \mathbb{R}^+$, τότε
 $\xi \in \mathbb{R}^+$

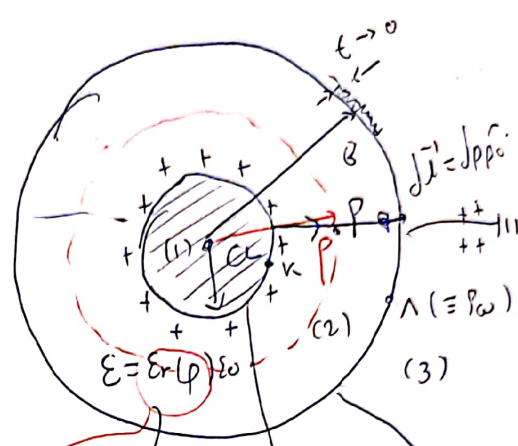
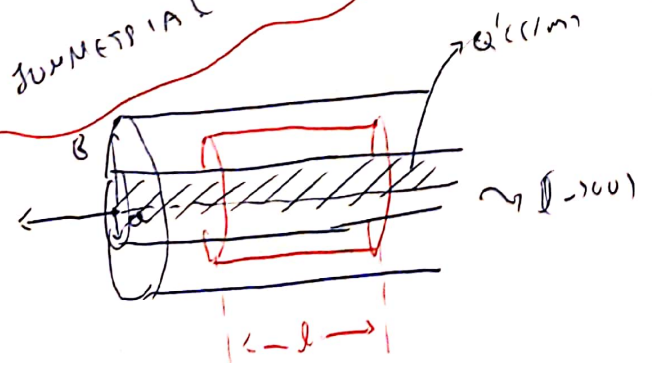
$$\begin{aligned} &= \varphi_n - \varphi_n^0 \\ &= \varphi_n = \sqrt{\gamma_0} \\ &\vec{E}_1 = \vec{0} \end{aligned}$$

|| Έτσι
 (cm) α το
 φορτίο τω.
 πυρήνα, and

το δάκτυλο

χω αν $\xi_r: \mathbb{R}^3 \rightarrow \mathbb{R}$,
 τότε $\xi: \mathbb{R}^3 \rightarrow \mathbb{R}$

η ύψωση των κελών
 του καλωδίου and το δάκτυλο μήκους l μήκους
 προς σύσφιξη. δε φέρει καίρια
 ΑΝΕΙΡΟ μήκος, άρα τέτοια καλωδίωση
 ΣΥΜΜΕΤΡΙΑ (που σημαίνει $l \rightarrow \infty$)



$$E_r(b) = 1.$$

η σφύρη διανομής
 σφαιρική του κέντρου (2)

μπαράειστος ή πρ P
 and τω ελίκω, άρα $E_r(r)$ (μη σφαιρική σφύρη)

$$\oint \vec{D}_2 \cdot d\vec{S} = Q_{\text{enclosed}} \rightarrow \int_0^l \int_0^{2\pi} \int_0^{2\pi} D_2(r) \hat{\rho}_0 \rho d\phi dz \hat{\rho} = Q' \cdot l \rightarrow$$

$$\rightarrow D_2(r) \cdot \rho \cdot 2\pi \cdot l = Q' \cdot l \rightarrow D_2(r) = \frac{Q'}{2\pi \rho} \hat{\rho}_0$$

$$\vec{E}_2(r) = \frac{Q'}{2\pi \epsilon} \frac{\hat{\rho}_0}{\rho}$$

$$\begin{aligned} E_r(r) &= ? \\ \text{ώστε } |\vec{E}_2(r)| &= \\ &= \sigma a l \end{aligned}$$

$$C' (F/m) = ?$$

(από το δάκτυλο μήκους
 χρησιμοποιείται)

Διασφύδα διακλινά στα
 άκρα τω καλωδίου.

$$V = \varphi_{\infty} - \varphi_1 = \varphi_2(a) - \varphi_2(b) = \frac{Q'}{2\pi\epsilon_0 b} (b-a) \sim \frac{Q'}{V} = C' (F/m) = \frac{2\pi\epsilon_0 b}{b-a} \quad -2-$$

$$\varphi_p = \int_p^{\infty} \vec{E} \cdot d\vec{x}' = \int_p^b \frac{Q'}{2\pi\epsilon r} \frac{1}{r_0} dp/p_0 = \frac{Q'}{2\pi} \int_p^b \frac{1}{\epsilon(r) \cdot \epsilon_0 \cdot p} dp = \frac{Q'}{2\pi\epsilon_0} \int_p^b \frac{1}{\epsilon(r) \cdot p} dp =$$

$$= \frac{Q'}{2\pi\epsilon_0} \int_p^b \frac{1}{p} dp = \frac{Q'}{2\pi\epsilon_0 b} \int_p^b \frac{1}{p/p} dp = \frac{Q'}{2\pi\epsilon_0 b} \int_p^b \frac{1}{p/p} dp =$$

$$\Rightarrow \varphi_2(p) = \frac{Q'}{2\pi\epsilon_0 b} (b-p), (v)$$

Άρα

$$C' = \frac{2\pi\epsilon_0 b}{b-a} (F/m)$$

$$\epsilon(p) = \frac{b}{p} \epsilon_0 (F/m)$$

$$\sim \text{Άρα } \epsilon_r(p) = \frac{Q'}{2\pi\epsilon_0 p \frac{Q'}{2\pi\epsilon_0 b}} = \frac{b}{p}$$

ορίστε $\epsilon_2(p) = C \leftarrow$

$$\frac{Q'}{2\pi\epsilon_r(p) \cdot \epsilon_0 \cdot p} = C \rightarrow$$

$$\rightarrow \epsilon_r(p) = \frac{Q'}{2\pi\epsilon_0 p \cdot C} = \frac{\kappa}{p}$$

$$\downarrow$$

$$\epsilon_r(b) = 1 \rightarrow$$

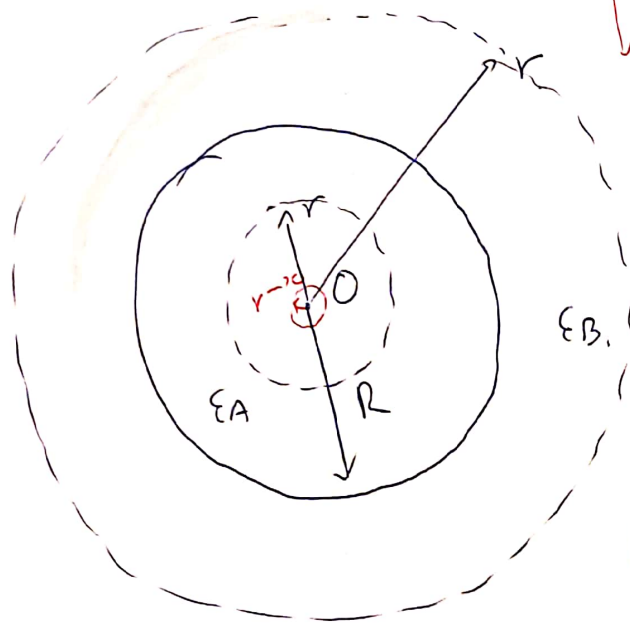
$$\rightarrow \frac{Q'}{2\pi\epsilon_0 \cdot b} = 1 \sim$$

$$\sim C = \frac{Q'}{2\pi\epsilon_0 b}$$

100N22
①:

Χώρος A: ($r \leq R$)

F, G γνωστά σκελετάς.



$$\varphi_A(r) = \frac{F \left(\frac{\epsilon}{m^3} \right)}{\epsilon_A \left(\frac{F}{m} \right)} \left[\frac{R^2 - r^2}{6} + \frac{R^2}{3} + \frac{G}{r} \right]$$

(v.)

Χώρος B: ($r > R$)

$$\varphi_B(r) = \frac{k R^3}{3 \epsilon_B r}$$

$1(v) = 1 \frac{(1)}{(F)}$

(α) $k = 1$ $\epsilon_A, \epsilon_B = 1$

(β) Δα τα φορτα κανον.

Γνωρίζουμε ότι $\varphi_A(R) = \varphi_B(R)$ hint
Συνεχώς

Από $G(m^3)$

Από $F(C/m^3)$

(Για τα επόμενα εργαζόμαστε με n/r

ισορροπία ΔΕΝ

έχει νόλο. Στο

εσωτερικό των διηλεκτρικών

-των ουραίων έχει

για χαρά)

$$\frac{F}{\epsilon_A} \left[\frac{R^2 - R^2}{6} + \frac{R^2}{3} + \frac{G}{R} \right] = \frac{k R^3}{3 \epsilon_B R} \rightarrow k = \frac{3 F \epsilon_B}{\epsilon_A R^2} \left[\frac{R^2}{3} + \frac{G}{R} \right] \Rightarrow \left[F \frac{\epsilon_B}{\epsilon_A} \left[1 + \frac{3 G}{R^3} \right] \right] = k$$

$$\vec{E}_A = -\nabla \varphi_A = -\frac{\partial \varphi_A}{\partial r} \hat{r}_0 = -\frac{\partial}{\partial r} \left(\frac{F}{\epsilon_A} \left(\frac{R^2 - r^2}{6} + \frac{R^2}{3} + \frac{G}{r} \right) \right) \hat{r}_0 = -\frac{F}{\epsilon_A} \left[-\frac{2r}{6} + 0 - \frac{G}{r^2} \right] \hat{r}_0$$

$$\vec{E}_A = \frac{F}{\epsilon_A} \left(\frac{r}{3} + \frac{G}{r^2} \right) \hat{r}_0 \quad (V/m)$$

$$\vec{E}_B = -\nabla \varphi_B = -\frac{\partial \varphi_B}{\partial r} \hat{r}_0 = -\frac{\partial}{\partial r} \left(\frac{k R^3}{3 \epsilon_B r} \right) \hat{r}_0 = \frac{k R^3}{3 \epsilon_B r^2} \hat{r}_0 \quad (V/m)$$

Για χωρίδια φορτίου $\leadsto \nabla \cdot \vec{D} = \rho_v \leadsto \underline{\underline{A}} \quad \rho_{vA} = \nabla \cdot \vec{D}_A = \nabla \cdot [\epsilon_A \vec{E}_A] = \epsilon_A \nabla \cdot \vec{E}_A =$

$$= \epsilon_A \cdot \frac{1}{r^2} \frac{\partial (r^2 E_A)}{\partial r} =$$

$$= \epsilon_A \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \cdot \frac{F}{\epsilon_A} \left(\frac{r}{3} + \frac{G}{r^2} \right) \right] =$$

$$= \cancel{\epsilon_A} \frac{1}{r^2} \frac{F}{\cancel{\epsilon_A}} \frac{\partial}{\partial r} \left(\frac{r^3}{3} + G \right) =$$

$$= \frac{F}{r^2} \left(\frac{3r^2}{3} + 0 \right) = F \leadsto$$

$$\leadsto \text{Άρα } \rho_{vA} = F \text{ (C/m}^3\text{)}$$

Το συνολικό φορτίο στη σφαίρα

$$\text{Επειδή } Q_A = \iiint_{V_A} \rho_{vA} dV_A =$$

$$= \iiint_0^R \int_0^\pi \int_0^{2\pi} F r^2 \sin \theta dr d\theta d\varphi = F \cdot \frac{4}{3} \pi R^3, \text{ (C)}$$

B

$$\rho_{vB} = \nabla \cdot \vec{D}_B = \nabla \cdot [\epsilon_B \vec{E}_B] = \epsilon_B \nabla \cdot \vec{E}_B =$$

$$= \epsilon_B \frac{1}{r^2} \frac{\partial (r^2 E_B)}{\partial r} = \epsilon_B \frac{1}{r^2} \frac{\partial}{\partial r} \left[\cancel{r^2} \cdot \frac{k R^3}{3 \epsilon_B \cancel{r^2}} \right] =$$

$$= 0, \text{ (C/m}^3\text{)}. \text{ Άρα } \vec{E}_B \text{ and } \vec{D}_B \text{ are zero everywhere.}$$

Για την επιφάνεια $\rightarrow n_{AB} \cdot (\vec{D}_B - \vec{D}_A) = \rho_s \mid_{\Delta \epsilon} \rightarrow$

$$\rightarrow n_{AB} \cdot [\vec{D}_B - \vec{D}_A] \mid_{r=R} = \rho_s \mid_{r=R} \rightarrow$$

$$\rightarrow \rho_s = \hat{r} \cdot \left[\cancel{\epsilon_B} \cdot \frac{k R^3}{3 \cancel{\epsilon_B} r^2} \cancel{r^2} - \cancel{\epsilon_A} \frac{F}{\cancel{\epsilon_A}} \left(\frac{r}{3} + \frac{G}{r^2} \right) \cancel{r^2} \right] \mid_{r=R} =$$

$$= \frac{k R^3}{3 R^2} - F \left(\frac{R}{3} + \frac{G}{R^2} \right) = \frac{k}{3} R - \frac{FR}{3} - \frac{FG}{R^2}$$

$$\hookrightarrow \text{όρα } \rho_s|_{r=R} = F \frac{\epsilon_B}{\epsilon_A} \left(1 + \frac{3G}{R^3} \right) \frac{R}{3} - \frac{FR}{3} - \frac{FG}{R^2} = \frac{FR}{3} \frac{\epsilon_B}{\epsilon_A} + \frac{FG}{R^2} \frac{\epsilon_B}{\epsilon_A} - \frac{FR}{3} - \frac{FG}{R^2} \sim$$

$$\sim \rho_s|_{r=R} = \frac{FR}{3} \left(\frac{\epsilon_B}{\epsilon_A} - 1 \right) + \frac{FG}{R^2} \left(\frac{\epsilon_B}{\epsilon_A} - 1 \right) = \left(\frac{\epsilon_B}{\epsilon_A} - 1 \right) \left(\frac{FR}{3} + \frac{FG}{R^2} \right), \text{ (C/m}^2\text{)}$$

$$Q_{\text{επιφανειακή}} = \iint_{\text{σφαίρα}} \rho_s dS = \iint_0^{2\pi} \int_0^\pi \left(\frac{\epsilon_B}{\epsilon_A} - 1 \right) \left(\frac{FR}{3} + \frac{FG}{R^2} \right) R^2 \sin\theta d\theta d\varphi \Big|_{r=R} = \left(\frac{\epsilon_B}{\epsilon_A} - 1 \right) \left(\frac{FR}{3} + \frac{FG}{R^2} \right) \cdot 4\pi R^2, \text{ (C)}$$

$\hookrightarrow$ αν δεν είχε σε κέντρο διασφαιρικό ή σφαιρική σύμμετρη, δε θα είχαμε αυτή την απλοποίηση.

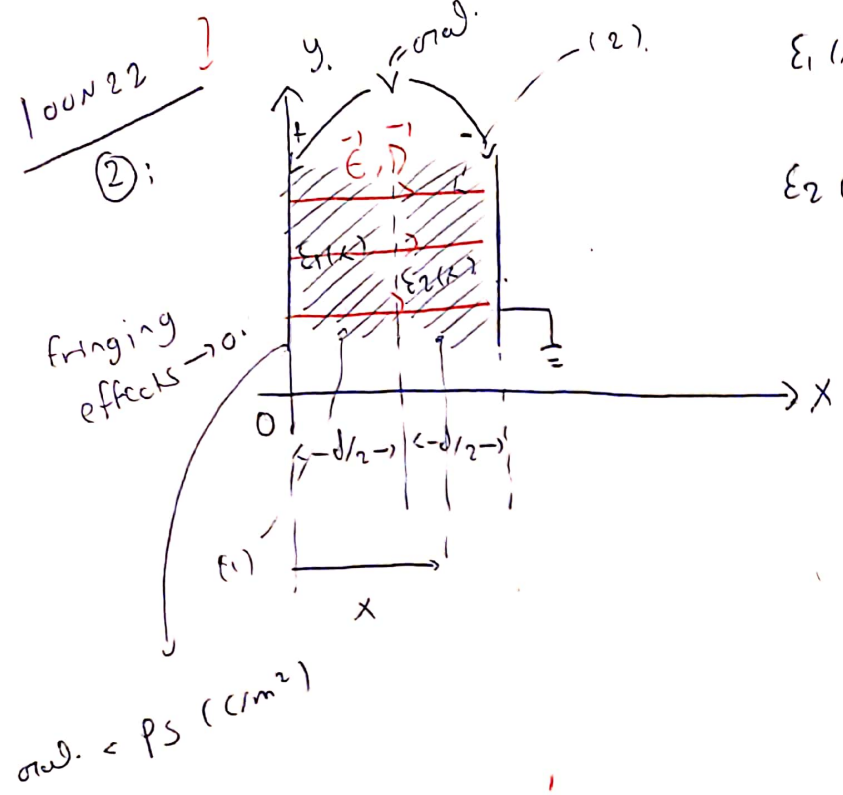
Για εσωτερικό φορτίο, ψάχνω μόνο στο κέντρο της σφαίρας $\leadsto q_{\text{επ.}} = \lim_{r \rightarrow 0} \oint_S \vec{D}_A \cdot d\vec{S} \rightarrow$ προς το κέντρο

$$\rightarrow q_{\text{επ.}} = \lim_{r \rightarrow 0} \iint_0^{2\pi} \int_0^\pi \cancel{\epsilon_A} \frac{F}{\cancel{\epsilon_A}} \left(\frac{r}{3} + \frac{G}{r^2} \right) r^2 \sin\theta d\theta d\varphi \cdot \cancel{r^2} =$$

$$= \lim_{r \rightarrow 0} \left[\cancel{\left[F \frac{r^3}{3} + FG \right]} \int_0^{2\pi} \cancel{d\varphi} \int_0^\pi \cancel{\sin\theta d\theta} \right] = FG \cdot 4\pi, \text{ (C)}$$

$\leadsto$ Άρα $q_{\text{επ.}} = 4\pi FG, \text{ (C)}$

100N22
(2):



$$\epsilon_1(x) = \epsilon_0 \left(1 + \frac{x}{d} \right)$$

$$\epsilon_2(x) = \epsilon_0 \left(2 - \frac{x}{d} \right)$$

$$(a) \vec{E}(x), \vec{D}(x), \varphi(x) = ?$$

Από τον νόμο του Gauss (φαινόμενα άκρων $\rightarrow 0$) θα έχουμε ορισμένα αποτελέσματα για το πεδίο x .

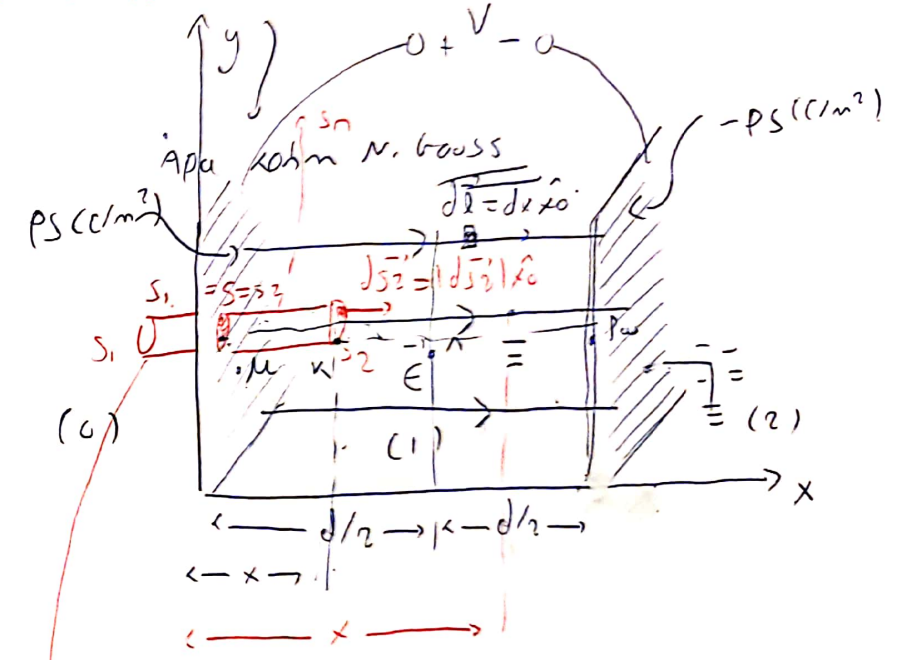
$$\iint_{S_2} D(x) \hat{x}_0 \cdot d\vec{S}_2 \hat{x}_0 = PS \cdot S \rightarrow$$

$$\rightarrow D(x) \iint_{S_2} dS_2 = PS \cdot S \sim D(x) = PS \sim$$

$$\sim \vec{D}(x) = PS \hat{x}_0 \text{ (C/m}^2\text{)}$$

$$\vec{E}_1(x) = \frac{\vec{D}(x)}{\epsilon_1(x)}$$

$$\vec{E}_2(x) = \frac{\vec{D}(x)}{\epsilon_2(x)}$$



$$\oint_S \vec{D} \cdot d\vec{S} = q_{\text{enclosed}}$$

$$\rightarrow \iint_{S_1} \vec{D} \cdot d\vec{S}_1 + \iint_{S_2} \vec{D} \cdot d\vec{S}_2 + \iint_{S_3} \vec{D} \cdot d\vec{S}_3 = q_{\text{enclosed}} \rightarrow$$

Επειδή οι δύο πλευρές είναι 0 οι γειγνόμενες αλληλεπιδράσεις $\rightarrow$

Επειδή οι δύο πλευρές S_1 και S_3 είναι παράλληλες και απέχουν πολύ, τότε $\vec{D} \perp d\vec{S}_1$ και $\vec{D} \perp d\vec{S}_3$.

$$\vec{D}(x) = \begin{cases} \epsilon_0 \hat{x}_0, & 0 \leq x \leq d/2 \\ \epsilon_0 \hat{x}_0, & d/2 \leq x \leq d. \end{cases}$$

$$\vec{E}(x) = \begin{cases} \frac{\rho_s d}{\epsilon_0 (x+d)} \hat{x}_0, & 0 \leq x \leq d/2 \\ \frac{\rho_s d}{\epsilon_0 (2d-x)} \hat{x}_0, & d/2 \leq x \leq d. \end{cases}$$

o.g. na $x=0$ $\leadsto n_{0 \rightarrow 1} \cdot (\vec{D}_1 - \vec{D}_0) = \rho_s |_{x=0} \Rightarrow$
 $\Rightarrow \rho_s |_{x=0} = \hat{x}_0 \cdot [\rho_s \hat{x}_0 - 0] = \rho_s \text{ (C/m}^2\text{)}$

o.g. na $x=d$ $\leadsto n_{1 \rightarrow 2} \cdot (\vec{D}_2 - \vec{D}_1) = \rho_s |_{x=d} \Rightarrow$
 $\Rightarrow \rho_s |_{x=d} = \hat{x}_0 (0 - \rho_s \hat{x}_0) = -\rho_s \text{ (C/m}^2\text{)}$

$$\hookrightarrow \varphi_1(x) = \frac{\rho_s d}{\epsilon_0} \left(\ln\left(\frac{3d/2}{x+d}\right) + \ln\left(\frac{3}{2}\right) \right)$$

$d/2 \leq x \leq d$

$$\varphi_{\equiv \rightarrow P_{\omega}} = \int_{\equiv}^{P_{\omega}} \vec{E}_2(x) d\vec{x} =$$

$$= \int_x^d \frac{\rho_s d}{\epsilon_0 (2d-x)} \hat{x}_0 dx \hat{x}_0 \quad \square$$

$$\varphi_{\equiv \rightarrow P_{\omega}} = \int_{\equiv}^{P_{\omega}} \vec{E}(x) d\vec{x} =$$

$$= \int_K^{\wedge} \vec{E}_1(x) \hat{x}_0 dx \hat{x}_0 + \int_{d/2}^{P_{\omega}} \vec{E}_2(x) \hat{x}_0 dx \hat{x}_0 =$$

$$= \int_x^{d/2} \frac{\rho_s d}{\epsilon_0 (x+d)} dx + \int_{d/2}^d \frac{\rho_s d}{\epsilon_0 (2d-x)} dx =$$

$$= \frac{\rho_s d}{\epsilon_0} \left[\ln(x+d) \right]_{x=d/2}^{d/2} + \frac{\rho_s d}{\epsilon_0} \left[-\ln(2d-x) \right]_{d/2}^d$$

$$= \frac{\rho_s d}{\epsilon_0} \left[\ln\left(\frac{3d/2}{x+d}\right) \right] + \frac{\rho_s d}{\epsilon_0} \left(\ln\left(\frac{3d/2}{d}\right) \right)$$

Προσοχή
 το double
 υπολογισμένο
 πάνω από
 τους μω
 ιδίως
 αντιστρέφω
 (ρω)

$$\int \frac{\rho_s d}{\epsilon_0} \left[-\ln(2d-x) \right]_x^d = \frac{\rho_s d}{\epsilon_0} \left(-\ln(2d-d) + \ln(2d-x) \right) = \frac{\rho_s d}{\epsilon_0} \left[\ln(2d-x) - \ln d \right] \Rightarrow$$

$$\Rightarrow \varphi_2(x) = \frac{\rho_s d}{\epsilon_0} \ln\left(\frac{2d-x}{d}\right) \rightsquigarrow \varphi(x) = \begin{cases} \frac{\rho_s d}{\epsilon_0} \left[\ln\left[\frac{3d/2}{x+d}\right] + \ln\left[\frac{3}{2}\right] \right] & 0 \leq x \leq d/2 \\ \frac{\rho_s d}{\epsilon_0} \ln\left(\frac{2d-x}{d}\right) & d/2 \leq x \leq d \end{cases}$$

$$\sim \text{d/ωs} \quad V = \varphi_{\text{max}} - \varphi_{\text{min}} = \varphi_1(0) - \varphi_2(d) \rightsquigarrow$$

$$\rightsquigarrow V = \frac{\rho_s d}{\epsilon_0} \ln(9/4) \rightsquigarrow \rho_s = \frac{\epsilon_0 V}{d \ln(9/4)} \text{ (C/m}^2\text{)}$$

$$C'' \text{ (F/m}^2\text{)} = \frac{\rho_s}{V} = \frac{\epsilon_0}{d \ln(9/4)} \text{ (F/m}^2\text{)}$$

$$\omega_e'' = \frac{1}{2} C'' V^2 = \frac{\epsilon_0}{2 d \ln(9/4)} \cdot V^2 \text{ (J/m}^2\text{)}$$

για εύρεση S των ηωων

$$\begin{aligned} C(F) &= C'' \cdot S \\ \omega_e(J) &= \omega_e'' \cdot S \end{aligned}$$

$$\hookrightarrow \varphi_1(d) = 0, \text{ γιατί αντιστοιχεί}$$

$$\varphi_2(d/2) = \frac{\rho_s d}{\epsilon_0} \ln\left(\frac{3}{2}\right), (V) \quad \left. \vphantom{\frac{\rho_s d}{\epsilon_0}} \right\} \rightarrow$$

$$\varphi_1(d/2) = \frac{\rho_s d}{\epsilon_0} \left[\ln\left(\frac{3}{2}\right) \right], (V)$$

$$\rightarrow \varphi_1(d/2) = \varphi_2(d/2), \text{ γιατί}$$

το δυνάμει

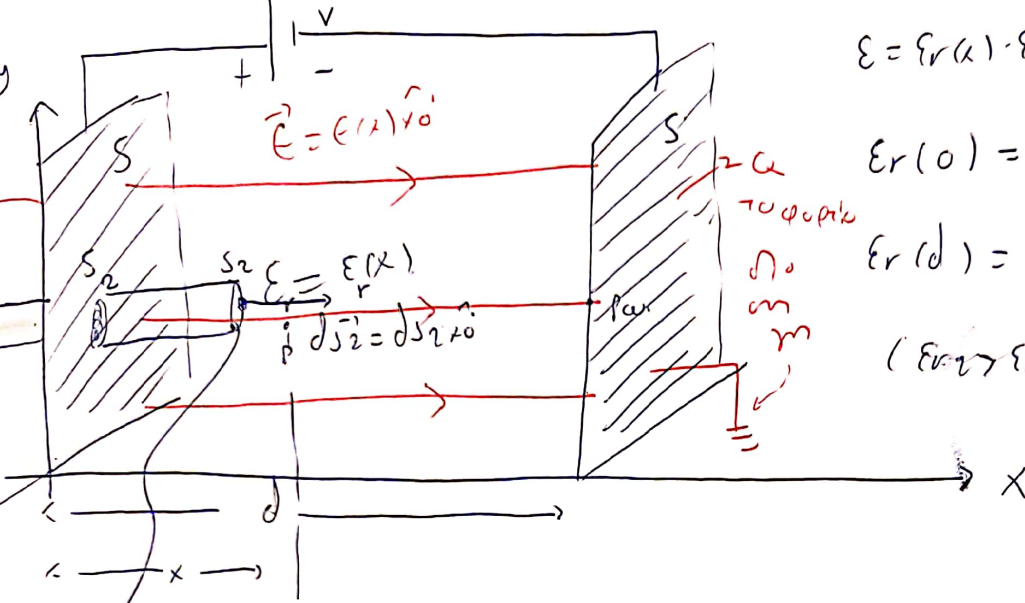
$$\varphi_1(0) = \frac{\rho_s d}{\epsilon_0} \left[\ln\left(\frac{3d/2}{d}\right) + \ln\left(\frac{2}{3}\right) \right] \overset{\text{για } x=d/2}{=} \frac{\rho_s d}{\epsilon_0} \left[\ln\left(\frac{3}{2}\right) + \ln\left(\frac{2}{3}\right) \right] =$$

$$\begin{aligned} &= \frac{\rho_s d}{\epsilon_0} \left[\ln\left(\frac{3}{2}\right) + \ln\left(\frac{2}{3}\right) \right] = \\ &= \frac{\rho_s d}{\epsilon_0} \ln\left(\frac{9}{4}\right), (V) \rightsquigarrow \end{aligned}$$

Σελ 22
 (4) (πρωτ): y

Σελω + α
 το φερτικό
 το ηλεκτρικό

$\vec{E} = 0$



$$\epsilon = \epsilon_r(x) \cdot \epsilon_0, \quad \epsilon_r(x) \sim \text{pollin} \text{ i } \text{Idrom} \text{ w } \text{pos } x$$

$$\epsilon_r(0) = \epsilon_{r1}$$

$$\epsilon_r(d) = \epsilon_{r2}$$

$$(\epsilon_{r2} > \epsilon_{r1})$$

$$\epsilon_r(x) - \epsilon_{r1} = \frac{\epsilon_{r2} - \epsilon_{r1}}{d - 0} (x - 0)$$

$$\epsilon_r(x) = \epsilon_{r1} + \frac{\epsilon_{r2} - \epsilon_{r1}}{d} \cdot x$$

pollin
 in Idrom tis stathis

$$\oint \vec{D} \cdot d\vec{S} = Q_{\text{free}} \rightarrow \int_{S_1} \vec{D}_1 \cdot d\vec{S}_1 + \int_{S_2} \vec{D}_2 \cdot d\vec{S}_2 + \int_{S_3} D(x) \hat{x}_0 \cdot d\vec{S}_3$$

and the surface x has D
 $P_S \cdot S_2 \rightarrow$ the surface

$$\rightarrow D(x) \cdot S_2 = P_S \cdot S_2 \sim D(x) = P_S \hat{x}_0 \quad (C/m^2)$$

$$\vec{E}(x) = \frac{P_S}{\epsilon(x)} \hat{x}_0 = \frac{P_S}{\epsilon_0 \left(\epsilon_{r1} + \frac{\epsilon_{r2} - \epsilon_{r1}}{d} \cdot x \right)} \hat{x}_0, \quad (V/m) \sim$$

$$\sim \varphi(x) = \int_P^{P_0} E \cdot d\vec{l} = \int_x^d \frac{P_S}{\epsilon_0 \left(\epsilon_{r1} + \frac{\epsilon_{r2} - \epsilon_{r1}}{d} \cdot x \right)} dx$$

$$\int \frac{1}{\frac{\epsilon_{r2} - \epsilon_{r1}}{d}} \ln \left(\epsilon_{r1} + \frac{\epsilon_{r2} - \epsilon_{r1}}{d} \cdot x \right) dx$$

$$\varphi(x) = \frac{\rho_s \cdot d}{\epsilon_0 (\epsilon_{r2} - \epsilon_{r1})} \cdot \ln \left[\frac{\epsilon_{r2}}{\epsilon_{r1} + \frac{\epsilon_{r2} - \epsilon_{r1}}{d} \cdot x} \right], (V)$$

$$\varphi(d) = 0 \quad (\text{προβλημα η διηλεκτρική ηγδυσία})$$

$$\varphi(c) = \varphi(c) - \varphi(d) = V$$

"γιατί η τιμή μπορεί να είναι και 2 ηγδυσ"

$$V = \frac{\rho_s d}{\epsilon_0 (\epsilon_{r2} - \epsilon_{r1})} \ln \left(\frac{\epsilon_{r2}}{\epsilon_{r1}} \right)$$

$$\hookrightarrow \rho_s = \frac{\epsilon_0 \cdot V (\epsilon_{r2} - \epsilon_{r1})}{d \ln \left(\frac{\epsilon_{r2}}{\epsilon_{r1}} \right)}$$

$$\rightarrow Q = \rho_s \cdot S, \quad C = \frac{Q}{V} \Rightarrow$$

$$\Rightarrow C = \frac{\epsilon_0 \cdot S (\epsilon_{r2} - \epsilon_{r1})}{d \ln \left(\frac{\epsilon_{r2}}{\epsilon_{r1}} \right)} \quad (F)$$

$$W_C = \frac{1}{2} C V^2, \quad (J)$$

$$\int \frac{1}{ax+b} dx = \frac{1}{a} \ln(ax+b)$$